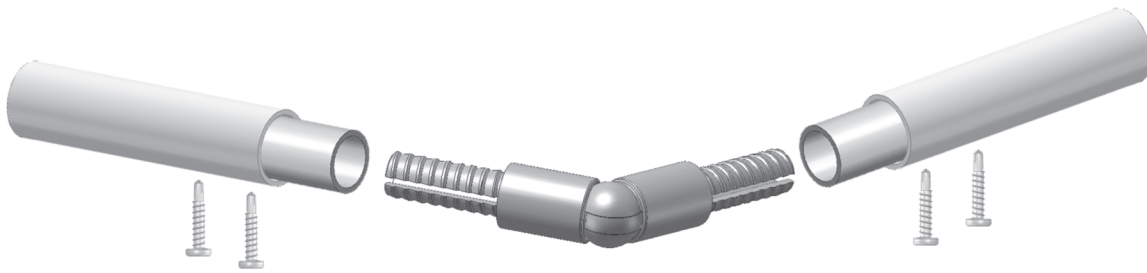


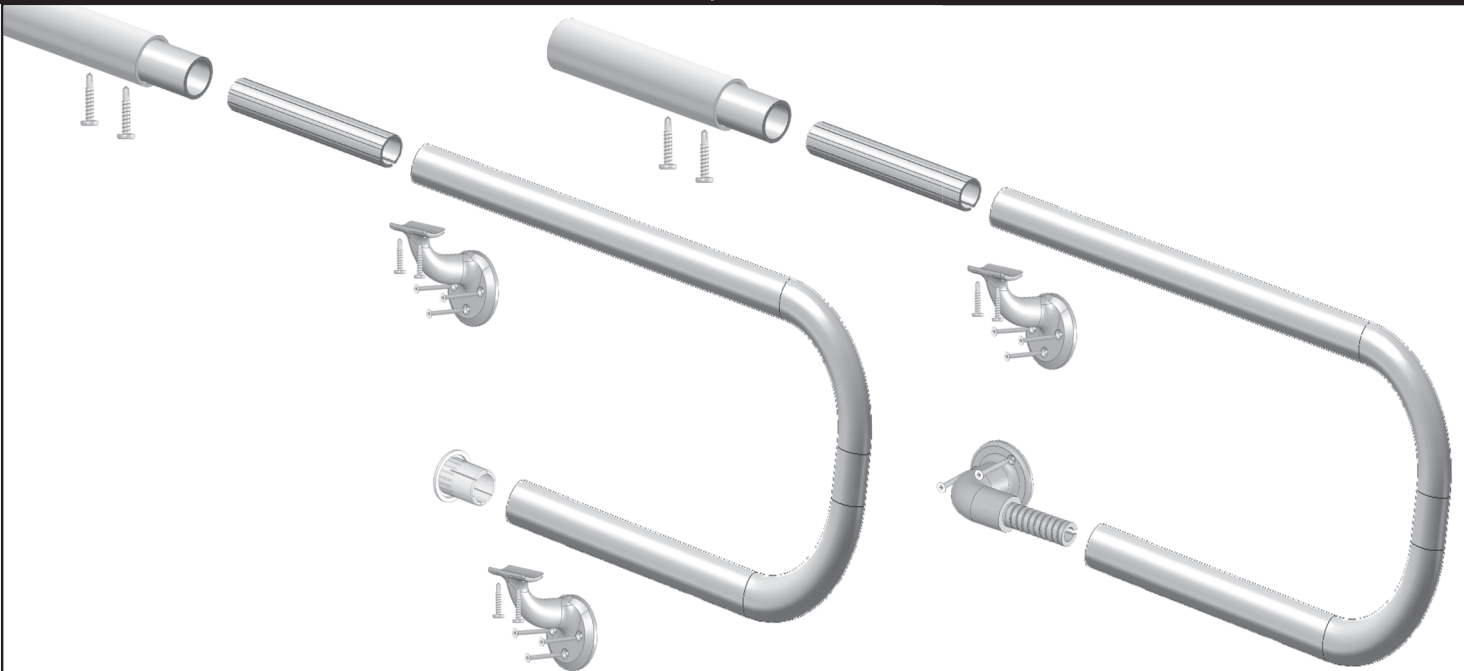
External Swivel - 0 to 60°



External Swivel Used for Step Degree Angles

- Adjust external swivel to desired angle and install set screw.
- Cut ADA tube with aluminum to desired length.
- Insert external swivel bracket into the ADA tube.
- Pre drill aluminum reinforced ADA tube and bracket with 5/32" drill bit and install 1" self tapping square drive screws.**

P-Loop Return



P-Loop Return is Used at the Bottom of Steps or Handicap Ramps

- Attach top bracket at desired location using appropriate connectors.
- Cut P-Loop to desired length.
- Connect P-Loop to ADA tube using the appropriate joiners (Straight, Adjustable joiners or External swivel).
- Set P-Loop on top of bracket and **pre drill with 5/32" drill bit and install 1" self tapping square drive screws.**
- Install desired lower bracket.

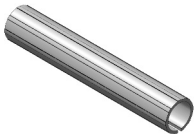
Typical ADA Railing Installation

Components:

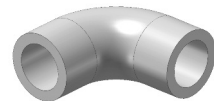
1 ADA Handrail w/Aluminum Insert



2 6" Straight Joiner



3 4" Inside/Outside Corner



4 Straight End Wall Bracket



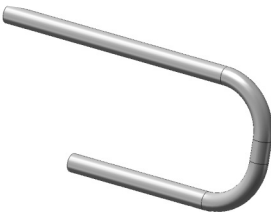
5 Inside Corner Bracket



6 End Cap



7 P-Loop Return



8 6" Adjustable Joiner



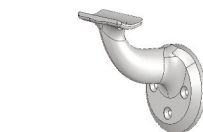
9 90° Quick Return Bracket



10 External Swivel



11 Stand Off Bracket



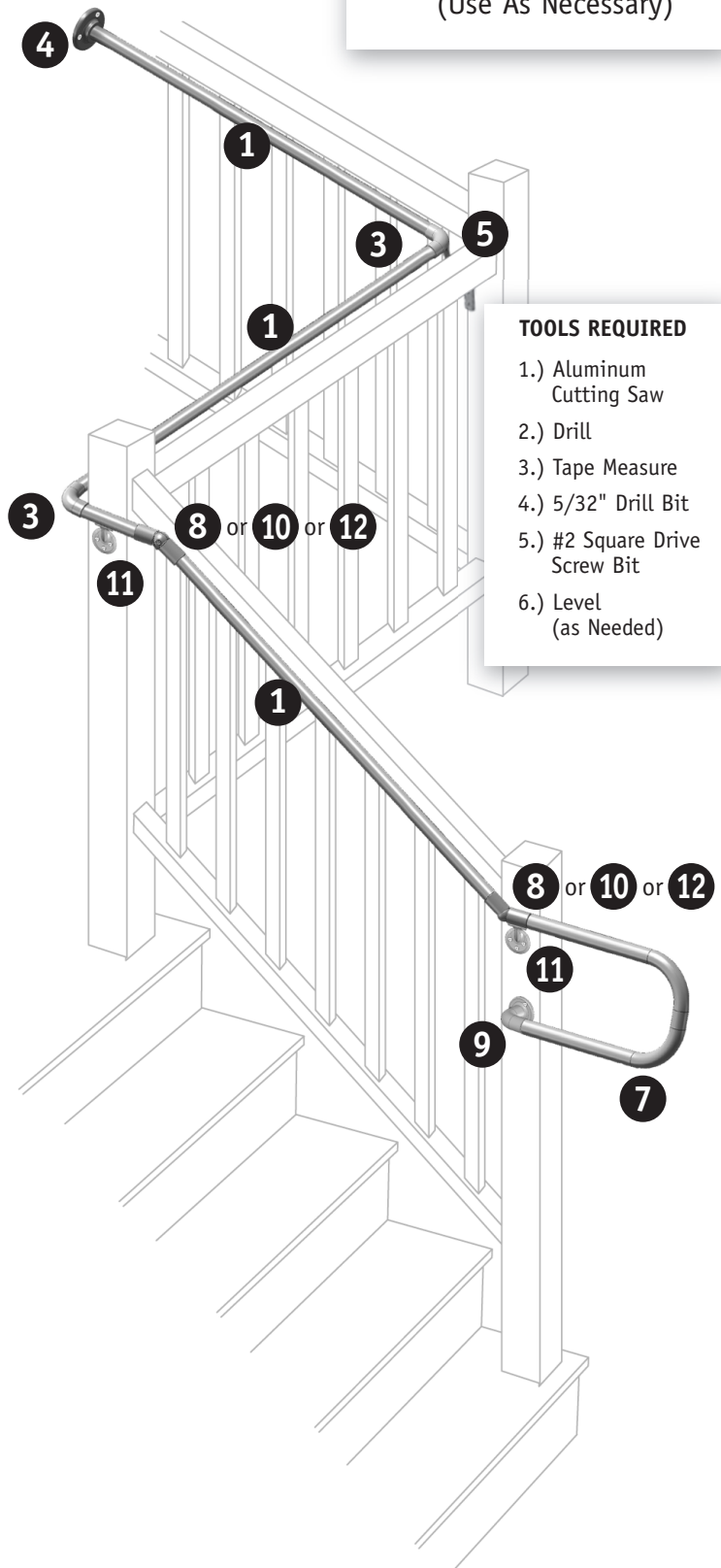
12 5°, 32°, or 38° Radius Bend



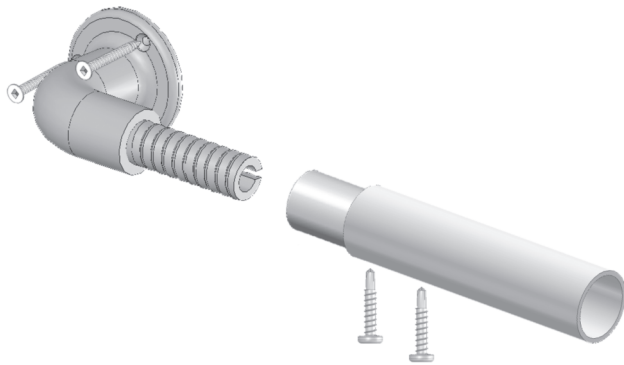
Optional Beauty Ring Joint Cover
(Use As Necessary)

TOOLS REQUIRED

- 1.) Aluminum Cutting Saw
- 2.) Drill
- 3.) Tape Measure
- 4.) 5/32" Drill Bit
- 5.) #2 Square Drive Screw Bit
- 6.) Level (as Needed)



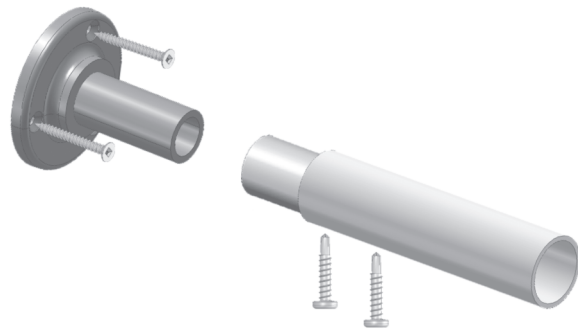
90° Quick Return Bracket



90 Degree Quick Return Bracket Used for Both Post and Wall Returns

- Cut ADA tube with aluminum to desired length.
- Insert 90 degree quick return bracket into the ADA tube.
- Pre drill and install 2 1/4" screws provided through the bracket.
- Pre drill aluminum reinforced tube and bracket with 5/32" drill bit and install 1" self tapping square drive screws.**

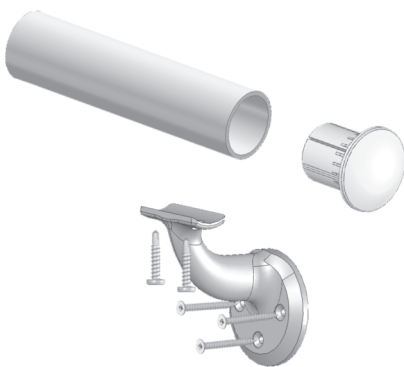
Straight End Wall Bracket



Straight End Wall Bracket Used for Straight Wall Connections

- Cut ADA tube with aluminum to desired length.
- Insert straight end bracket into the ADA tube.
- Pre drill and install 2 1/4" screws provided through the bracket.
- Pre drill aluminum reinforced tube and bracket with 5/32" drill bit and install 1" self tapping square drive screws.**

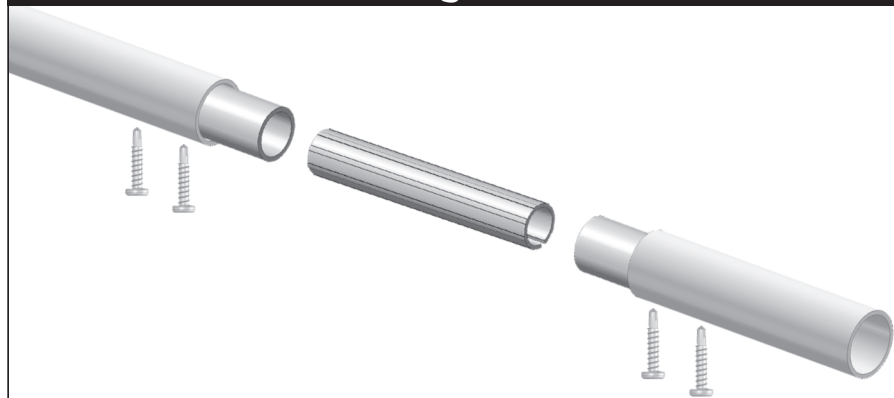
Stand Off Bracket



Stand Off Bracket Used for Rail Applications on Walls or Posts

- Pre drill and install 2 1/4" square drive screws provided through the bracket into the mounting surface.
- Place rail at appropriate location on the bracket.
- Pre drill aluminum reinforced ADA tube with 5/32" drill bit and install 1" self tapping square drive screws.**
- Place end cap on rail.

6" Straight Joiner



Internal 6" Straight Joiner Used to Connect 2 ADA Tubes or 4" Inside/Outside Corner

- Cut ADA tube with aluminum to desired length.
- Pre drill aluminum reinforced ADA tube and joiner with 5/32" drill bit and install 1" self tapping square drive screws.**

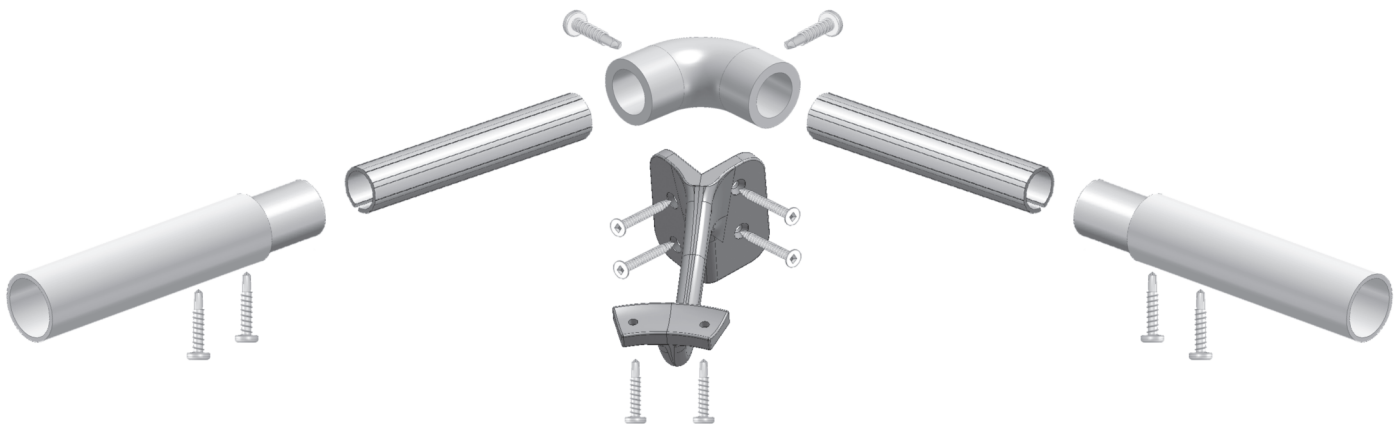
6" Adjustable Joiner - 0 to 90°



Internal 6" Adjustable Joiner Used for Degree Angles

- a) Miter cut ADA tube with aluminum to desired length.
- b) Install adjustable joiner inside both ends of the tube.
- b) **Pre drill ADA tube with 5/32" drill bit and install 1" self tapping square drive screws.**

4" Inside/Outside Corner with Inside Corner Bracket



4" Corner Can Be Used for Both Inside and Outside Corners

- a) Install 2 straight joiners into the 4" inside/outside corner.
- b) Pre drill 4" inside/outside corner and straight joiner with drill bit and install self tapping screw provided.
- c) Cut ADA tube with aluminum to desired length.
- d) **Pre drill aluminum reinforced ADA tube and joiner with 5/32" drill bit and install 1" self tapping square drive screws.**

4" Inside/Outside Corner Bracket

- a) Pre drill and install 2 1/4" square drive screws screws provided through the bracket into the mounting surface (corner of post).
- b) Place 4" inside/outside corner at appropriate location on the bracket.
- c) **Pre drill 4" inside/outside corner with 5/32" drill bit and install 1" self tapping square drive screws.**